

Systematic approach to sonographic evaluation of the pelvis in women with suspected endometriosis, including terms, definitions and measurements: a consensus opinion from the International Deep Endometriosis Analysis (IDEA) group

S. GUERRIERO¹#, G. CONDOUS²#, T. VAN DEN BOSCH³, L. VALENTIN⁴, F. P. G. LEONE⁵, D. VAN SCHOUBROECK³, C. EXACOSTOS⁶, A. J. F. INSTALLÉ⁷, W. P. MARTINS⁸, M. S. ABRAO⁹, G. HUDELIST¹⁰, M. BAZOT¹¹, J. L. ALCAZAR¹², M. O. GONÇALVES¹³, M. A. PASCUAL¹⁴, S. AJOSSA¹, L. SAVELLI¹⁵, R. DUNHAM¹⁶, S. REID¹⁷, U. MENAKAYA¹⁸, T. BOURNE¹⁹, S. FERRERO²⁰, M. LEON²¹, T. BIGNARDI²², T. HOLLAND²³, D. JURKOVIC²³, B. BENACERRAF²⁴, Y. OSUGA²⁵, E. SOMIGLIANA²⁶ and D. TIMMERMAN³

¹Department of Obstetrics and Gynecology, University of Cagliari, Policlinico Universitario Duilio Casula, Monserrato, Cagliari, Italy; ²Acute Gynaecology, Early Pregnancy & Advanced Endosurgery Unit, Sydney Medical School Nepean, University of Sydney, Nepean Hospital, Penrith, NSW, Australia; ³Department Obstetrics and Gynecology, University Hospitals, KU Leuven, Leuven, Belgium and Department of Obstetrics and Gynecology, Tienen Regional Hospital, Tienen, Belgium; ⁴Lund University, Department of Obstetrics and Gynecology, Skåne University Hospital, Malmö, Sweden; ⁵Department of Obstetrics and Gynecology, Clinical Sciences Institute L. Sacco, Milan, Italy; ⁶Department of Biomedicine and Prevention, Obstetrics and Gynecological Clinic, Faculty of Medicine, University of Rome 'Tor Vergata', Rome, Italy and Ospedale Generale S. Giovanni Calibita Fatebene Fratelli, Rome, Italy; ⁷KU Leuven, Department of Electrical Engineering (ESAT), STADIUS, Center for Dynamical Systems, Signal Processing and Data Analytics, Leuven, Belgium and iMinds Medical IT, Leuven, Belgium; ⁸Department of Obstetrics and Gynecology, Ribeirão Preto Medical School, University of São Paulo, São Paulo, Brazil; ⁹Endometriosis Division, Obstetrics and Gynecological Department, São Paulo University, São Paulo, Brazil; ¹⁰Hospital St John of God Johannes, Vienna, Austria; ¹¹Department of Radiology, Hôpital Tenon, Assistance Publique-Hôpitaux de Paris and Université Pierre et Marie Curie, Paris, France; ¹²Department of Obstetrics and Gynecology, Clínica Universidad de Navarra School of Medicine, University of Navarra, Pamplona, Spain; ¹³Clinica Medicina da Mulher and RDO Medicina Diagnóstica, São Paulo, Brazil; ¹⁴Department of Obstetrics, Gynaecology and Reproduction, Institut Universitari Dexeus, Barcelona, Spain; ¹⁵Gynecology and Early Pregnancy Ultrasound Unit, S. Orsola - Malpighi Hospital, University of Bologna, Bologna, Italy; ¹⁶Department of Radiology, Leeds Teaching Hospitals NHS Trust, Leeds, UK; ¹⁷Department of Obstetrics and Gynaecology, Wollongong Hospital, Wollongong, NSW, Australia; ¹⁸Department of Obstetrics and Gynaecology Calvary Public Hospital & JUNC Specialist Imaging & Women's Center, Canberra, Australia; ¹⁹Queen Charlotte's & Chelsea Hospital, Imperial College, London, UK; ²⁰Unit of Obstetrics and Gynaecology, IRCCS AOU San Martino - IST, Genova, Italy and Department of Neurosciences, Rehabilitation, Ophthalmology, Genetics, Maternal and Child Health (DiNOGMI), University of Genova, Genova, Italy; ²¹Ultrasound Unit, Department of Gynaecology and Obstetrics, Clínica Indisa, Santiago, Chile; ²²Department of Obstetrics and Gynecology, Azienda, Ospedaliera Niguarda Ca' Granda, Milan, Italy; ²³Institute for Women's Health, University College Hospital, London, UK; ²⁴Department of Obstetrics, Gynecology and Radiology, Brigham and Women's Hospital and Harvard Medical School, Boston, MA, USA; ²⁵Department of Obstetrics and Gynecology, The University of Tokyo, Tokyo, Japan; ²⁶Fondazione Ca' Granda, Ospedale Maggiore Policlinico, Milan, Italy

KEYWORDS: anterior and posterior compartments; deep infiltrating endometriosis; endometrioma; laparoscopy; ultrasound

ABSTRACT

The IDEA (International Deep Endometriosis Analysis group) statement is a consensus opinion on terms, definitions and measurements that may be used to describe the sonographic features of the different phenotypes of endometriosis. Currently, it is difficult to compare results between published studies because authors use different terms when describing the same structures and anatomical locations. We hope that the terms and definitions

suggested herein will be adopted in centers around the world. This would result in consistent use of nomenclature when describing the ultrasound location and extent of endometriosis. We believe that the standardization of terminology will allow meaningful comparisons between future studies in women with an ultrasound diagnosis of endometriosis and should facilitate multicenter research. Copyright © 2016 ISUOG. Published by John Wiley & Sons Ltd.

Correspondence to: Prof. G. Condous, Acute Gynaecology, Early Pregnancy & Advanced Endosurgery Unit, Sydney Medical School Nepean, University of Sydney, Nepean Hospital, Penrith, New South Wales, Australia (e-mail: gcondous@omnigynaecare.com.au)

#S.G. and G.C. are joint first authors.

Accepted: 25 April 2016

INTRODUCTION

Endometriosis is a common gynecological problem, affecting approximately 5% of women¹. The disease can be found in many sites throughout the pelvis, in particular the ovaries, pelvic peritoneum, pouch of Douglas (POD), rectum, rectosigmoid, rectovaginal septum (RVS), uterosacral ligaments (USLs), vagina and urinary bladder. Correct site-specific diagnosis is fundamental in defining the optimal treatment strategy for endometriosis. Non-invasive imaging methods are required to map accurately the location and extent of endometriotic lesions. The recent consensus statement produced by the World Endometriosis Society recommended the establishment of centers of expertise for the management of higher-stage disease². This recommendation requires a reliable pre-operative system of triage which enables immediate understanding of the location and severity of disease.

Increasingly, endometriosis is being managed medically and surgery can be avoided or delayed in a growing proportion of cases. Transvaginal sonography (TVS) is the first-line imaging technique in the diagnosis of pelvic endometriosis and in particular for deep infiltrating endometriosis (DIE)³. It is important to note, however, that there is substantial heterogeneity in the reported sensitivity and specificity of TVS with regard to detection of DIE, irrespective of its location^{4,5}. Adding ultrasound examination by an experienced operator to history and pelvic examination improves the accuracy of diagnosis of pelvic endometriosis^{6,7}. In their meta-analysis, Hudelist *et al.*⁸ concluded that TVS with or without the use of prior bowel preparation is an accurate test for non-invasive, presurgical detection of DIE of the rectosigmoid. Although the diagnostic performance of ultrasound for detecting DIE reported by individual units is excellent for certain anatomical locations^{9–11}, the lack of standardized definitions in the sonographic classification and diagnosis of DIE is a general cause for concern. This lack of uniformity when classifying anatomical location and extent of disease contributes to the considerable variation in the reported diagnostic accuracy of TVS in the diagnosis of endometriosis.

The aim of this consensus opinion is to ensure that the ultrasound examination of a woman with potentially underlying endometriosis is performed in a standardized manner, that the measurement of endometriotic lesions is standardized and that the terminology used when describing the location of DIE and the sonographic features of DIE and other manifestations of endometriosis (endometriomas, adenomyosis, pelvic adhesions) is uniform. This consensus opinion should be useful in clinical practice as well as in research. We believe that careful definition of ultrasound-detected DIE will facilitate interpretation of research and lead to improved clinical care.

METHODS

This work is based on the opinion of a panel of clinicians, gynecological sonologists, advanced

laparoscopic surgeons and radiologists (International Deep Endometriosis Analysis (IDEA) group) with expertise in diagnosis and management of endometriosis. Criteria used to invite the experts to participate in this consensus process included their having significant peer-reviewed publications in the field of diagnosis and management of endometriosis. An initial statement was presented in 2011 at the ISUOG congress in Copenhagen¹², incorporating several suggestions from all participants. A first draft was written in December 2014 by a joint effort of the two first authors (S.G. and G.C.) and sent to all coauthors. All coauthors had the opportunity to comment within a fixed time limit. Reply was mandatory for coauthorship. Taking all comments into account, a revised draft was then sent to all coauthors. In case of conflicting opinions, a consensus was proposed after discussion between the two first authors and the last author (D.T.). This pathway was repeated until a consensus between all authors was reached. The consensus also included ultrasound images/videos and schematic drawings to illustrate the text. After 13 revisions, the manuscript was deemed ready for submission.

In addition to terms, definitions and measurements to describe the sonographic features of DIE, adhesions, adenomyosis and endometriomas, this consensus opinion includes recommendations regarding how to take a history, how to perform a clinical examination, how to perform an ultrasound examination and which ultrasound modality to use when examining patients with suspected or known endometriosis. DIE anatomical locations in this consensus were modified from Chapron's anatomical distribution of pelvic DIE¹³.

HISTORY

A detailed clinical history should be taken for all women with suspected endometriosis, with particular emphasis on symptoms which could be attributed to endometriosis^{14,15}. The following should be noted specifically: age; height; weight; ethnic origin; parity; bleeding pattern (regular, irregular or absent); last menstrual period; previous surgery for endometriosis (type, effect); previous myomectomy or Cesarean delivery (these entail increased risk of DIE in the bladder); family history of endometriosis; previous non-surgical treatment for endometriosis (type, duration, effect); subfertility including duration of subfertility; treatment for infertility and outcome of fertility treatment; pain (dysmenorrhea, dyspareunia, dysuria, dyschezia, chronic pelvic pain); hematochezia and/or hematuria. The onset and duration of symptoms should be noted and, if possible, the intensity of the pain recorded by letting the patient use a visual analog scale or investigating it with a 0–10 narrative numeric rating scale.

PELVIC EXAMINATION

A pelvic examination should be performed either before or after the pelvic ultrasound scan, with the aim of defining

the presence or absence of vaginal and/or low rectal endometriosis⁷. The pelvic examination should include speculum examination (direct visualization of vaginal or cervical DIE) and vaginal palpation. Mobility, fixation and/or tenderness of the uterus should be evaluated carefully. Site-specific tenderness in the pelvis should also be evaluated.

SONOGRAPHY OVERVIEW

The purpose of performing an ultrasound examination in a woman with suspected endometriosis is to try to explain underlying symptoms, map the disease location and assess the severity of disease prior to medical therapy or surgical intervention. Various ultrasound approaches have been published, but to date none has been externally validated^{16,17}. We propose four basic sonographic steps when examining women with suspected or known endometriosis, as shown in Figure 1. Note that these steps can be adopted in this or any order as long as ALL four steps are performed to confirm/exclude the different forms of endometriosis.

Using TVS as the first-line imaging tool, the operator should examine the uterus and the adnexa. The mobility of the uterus should be evaluated: normal, reduced or fixed ('question mark sign')¹⁸. Sonographic signs of adenomyosis should be searched for and described using the terms and definitions published in the Morphological Uterus Sonographic Assessment consensus opinion¹⁹.

The presence or absence of endometriomas (Figure S1a), their size, measured systematically in three orthogonal planes (see 'Measurement of lesions', below), the number of endometriomas and their ultrasound appearance should be noted²⁰. The sonographic characteristics of any endometrioma should be described using the International Ovarian Tumor Analysis terminology²¹. An atypical endometrioma (Figure S1b) is defined as a unilocular-solid

mass with ground glass echogenicity with a papillary projection, a color score of 1 or 2 and no flow inside the papillary projection²⁰.

Ovarian endometriomas are associated frequently with other endometriotic lesions, such as adhesions and DIE^{22,23}. The 'kissing' ovaries sign (Figure S2) suggests that there are severe pelvic adhesions; bowel and Fallopian tube endometriosis are significantly more frequent in women with kissing ovaries *vs* those without kissing ovaries: 18.5% *vs* 2.5% and 92.6% *vs* 33%, respectively²⁴.

Endometriomas may undergo decidualization in pregnancy, in which case they can be confused with an ovarian malignancy on ultrasound examination (Figure S3)²⁵. Simultaneous presence of other endometriotic lesions may facilitate a correct diagnosis of endometrioma in pregnancy and minimize the risk of unnecessary surgery.

The second step is to search for sonographic 'soft markers', i.e. site-specific tenderness (SST) and fixed ovaries. The presence of soft markers increases the likelihood of superficial endometriosis and adhesions^{26,27}. By applying pressure between the uterus and ovary, one can assess if the ovary is fixed to the uterus medially, to the pelvic side wall laterally or to the USLs. The presence of adhesions can also be suspected if, on palpation with the probe and/or abdominal palpation with the free hand, the ovaries or the uterus appear to be fixed to adjacent structures (broad ligament, POD, bladder, rectum and/or parietal peritoneum). If there is pelvic fluid, fine strands of tissue (adhesions) may be seen between the ovary (with or without endometrioma) and the uterus or the peritoneum of the POD^{27–30}.

If there are endometriomas or pelvic endometriosis, the Fallopian tubes are frequently involved in the disease process. Adhesions may distort the normal Fallopian tubal course and occlusion of the Fallopian tube(s) by endometriotic foci or distal tubular adhesions may also occur. As a consequence, a sactosalpinx may develop. For these reasons, hydrosalpinx/hematosalpinx and peritoneal cysts should be searched for and reported.

The third step is to assess the status of the POD using the real-time TVS-based 'sliding sign'. In order to assess the sliding sign when the uterus is anteverted (Figure 2a), gentle pressure is placed against the cervix using the transvaginal probe, to establish whether the anterior rectum glides freely across the posterior aspect of the cervix (retrocervical region) and posterior vaginal wall. If the anterior rectal wall does so, the 'sliding sign' is considered positive for this location (Videoclip S1a). The examiner then places one hand over the woman's lower anterior abdominal wall in order to ballot the uterus between the palpating hand and the transvaginal probe (which is held in the other hand), to assess whether the anterior bowel glides freely over the posterior aspect of the upper uterus/fundus. If it does so, the sliding sign is also considered positive in this region (Videoclip S1b). When the sliding sign is found to be positive in both of these anatomical regions (retrocervix and posterior uterine fundus), the POD is recorded as being not obliterated.

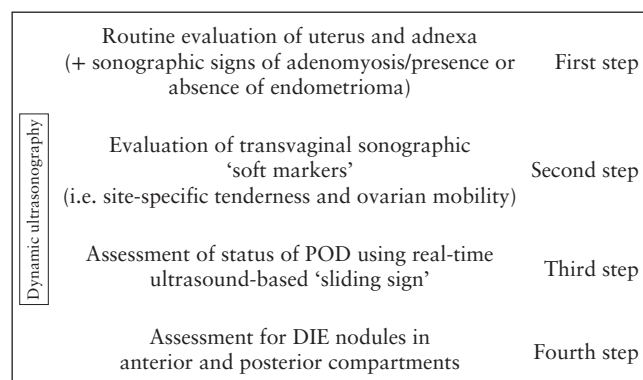


Figure 1 Four basic sonographic steps for examining women with clinical suspicion of deep infiltrating endometriosis (DIE) or known endometriosis. All steps should be performed, but not necessarily in this order. Note, bladder should contain small amount of urine. Dynamic ultrasonography is when the operator performing the ultrasound examination assesses both the pelvic organs and their mobility in real-time. POD, pouch of Douglas.

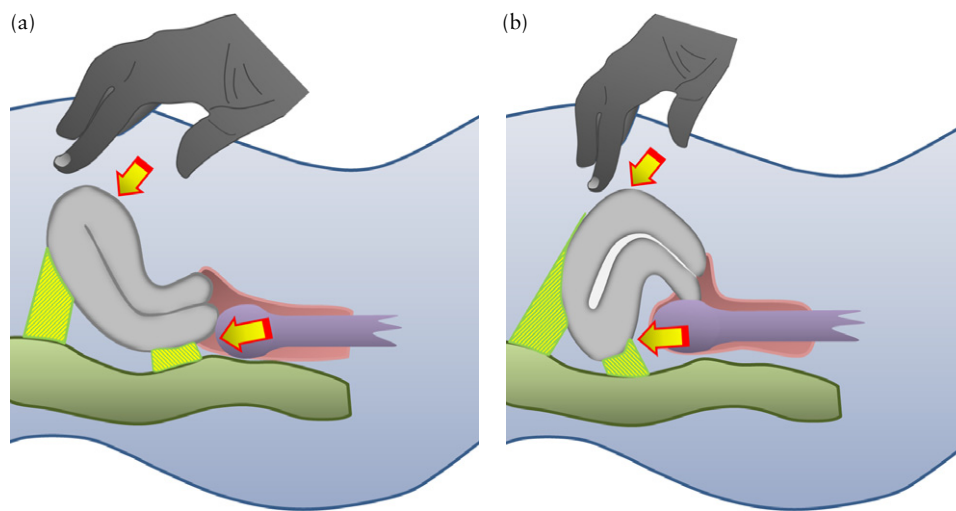


Figure 2 Schematic drawings demonstrating how to elicit the 'sliding sign' in an anteverted uterus (a) and a retroverted uterus (b).

If on TVS it is demonstrated that either the anterior rectal wall or the anterior sigmoid wall does not glide smoothly over the retrocervix or the posterior uterine fundus, respectively, i.e. at least one of the locations has a negative sliding sign, then the POD is recorded as obliterated^{31,32}.

Demonstrating and describing the real-time ultrasound-based sliding sign in a retroverted uterus is different (Figure 2b). Gentle pressure is placed against the posterior upper uterine fundus with the transvaginal probe, to establish whether the anterior rectum glides freely across the posterior upper uterine fundus. If the anterior rectum does so, the sliding sign is considered to be positive for this location (Videoclip S2a). The examiner then places one hand over the woman's lower anterior abdominal wall in order to ballot the uterus between the palpating hand and transvaginal probe (which is held in the other hand), to assess whether the anterior sigmoid glides freely over the anterior lower uterine segment. If it does so, the sliding sign is also considered to be positive in this region (Videoclip S2b). As long as the sliding sign is found to be positive in both of these anatomical regions (i.e. the posterior uterine fundus and the anterior lower uterine segment), the POD is recorded as non-obliterated³³.

The fourth step is to search for DIE nodules in the anterior and posterior compartments. To assess the anterior compartment, the transducer is positioned in the anterior fornix of the vagina. If bladder endometriosis is suspected on the basis of symptoms, patients should be asked not to empty their bladder completely before the ultrasound examination. A slightly filled bladder facilitates evaluation of the walls of the bladder and detection and description of endometriotic nodules. Finally, the transducer is positioned in the posterior fornix of the vagina and slowly withdrawn through the vagina to allow visualization of the posterior compartment. Some authors advocate the use of bowel preparation on the evening before the pelvic scan and the use of a rectal enema within an hour before the ultrasound

examination to eliminate fecal residue and gas in the rectosigmoid^{34–37}. However, this is not mandatory, and there are no published prospective studies comparing TVS with and without bowel preparation for the diagnosis of bowel DIE. In a recent meta-analysis, TVS, either with or without bowel preparation, was found to be an accurate predictor of rectosigmoid DIE⁸.

COMPARTMENTAL EVALUATION

Anterior compartment

The anterior compartment includes the following anatomical locations: urinary bladder, uterovesical region and ureters.

Urinary bladder

Bladder DIE occurs more frequently in the bladder base and bladder dome than in the extra-abdominal bladder (Videoclip S3)³⁸. The bladder is best scanned if it contains a small amount of urine because this reduces false-negative findings. Although Savelli *et al.*³⁸ described two zones (bladder base and dome), we propose dividing the bladder ultrasound assessment into four zones (Figure 3): (i) the trigonal zone, which lies within 3 cm of the urethral opening, is a smooth triangular region delimited by the two ureteral orifices and the internal urethral orifice (Figure S4a); (ii) the bladder base, which faces backward and downward and lies adjacent to both the vagina and the supravaginal cervix (Figure S4b); (iii) the bladder dome, which lies superior to the base and is intra-abdominal (Figure S4c); and (iv) the extra-abdominal bladder (Figure S4d). Figure S5 and Videoclip S3 demonstrate the most frequent location of endometriotic bladder nodules, i.e. the bladder base.

On two-dimensional (2D) ultrasound the appearance of DIE in the anterior compartment can be varied, including hypoechoic linear or spherical lesions, with or without regular contours involving the muscularis

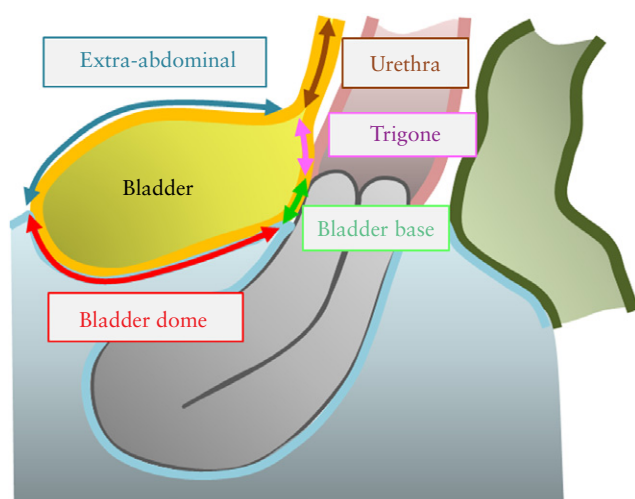


Figure 3 Schematic drawing illustrating the four bladder zones: trigone, bladder base, bladder dome and extra-abdominal bladder. The demarcation point between the base and the dome of the bladder is the uterovesical pouch.

(most common) or (sub)mucosa of the bladder^{6,38–43}. The dimensions of the bladder nodule should be measured in three orthogonal planes. Bladder DIE is diagnosed only if the muscularis of the bladder wall is affected; lesions involving only the serosa represent superficial disease.

Uterovesical region

Obliteration of the uterovesical region can be evaluated using the sliding sign, i.e. the transvaginal probe is placed in the anterior fornix and the uterus is balloted between the probe and one hand of the operator placed over the suprapubic region. If the posterior bladder slides freely over the anterior uterine wall, then the sliding sign is positive and the uterovesical region is classified as non-obiterated (Videoclip S4). If the bladder does not slide freely over the anterior uterine wall, then the sliding sign is negative and the uterovesical region is classified as obliterated⁴⁴ (Figure S6). Adhesions in the anterior pelvic compartment are present in nearly one third of women with a previous Cesarean section and are not necessarily a sign of pelvic endometriosis⁴⁴.

Ureters

The distal ureters should be examined routinely using the transvaginal probe. The ureters can be found by identifying the urethra in the sagittal plane and moving the probe towards the lateral pelvic wall. The intravesical segment of the ureter is identified and its course followed to where it leaves the bladder and then further, to the pelvic side wall and up to the level of the bifurcation of the common iliac vessels. It is helpful to wait for peristalsis to occur as this confirms ureteric patency. Ureters typically appear as long tubular hypoechoic structures, with a thick hyperechoic mantle, extending from the lateral aspect of the bladder base towards the common iliac vessels. Dilatation of the ureter due to endometriosis is caused by stricture (from

either extrinsic compression or intrinsic infiltration) and the distance from the distal ureteric orifice to the stricture should be measured (Figure S7)^{35,45,46}. Thorough evaluation of the ureter at the time of surgery is important in all cases in which ureteral involvement is suspected.

In all women with DIE, a transabdominal scan of the kidney to search for ureteral stenosis is necessary, because the prevalence of endometriotic lesions in the urinary tract may be underestimated and women with DIE involving the ureter may be asymptomatic^{47–51}. The degree of hydronephrosis should be assessed and graded using generally accepted ultrasound criteria⁵². Women with evidence of hydronephrosis should be referred for urgent stenting of a stenosed ureter to prevent further loss of renal function.

Posterior compartment

According to Chapron *et al.*⁵³, the most common sites of DIE in the posterior compartment are: USLs, posterior vaginal fornix, anterior rectum/anterior rectosigmoid junction and sigmoid colon. Sonographic assessment of the posterior compartment should aim at identifying the number, size and anatomical location of DIE nodules affecting these structures. On TVS, posterior compartment DIE lesions appear as hypoechoic thickening of the wall of the bowel or vagina, or as hypoechoic solid nodules which may vary in size and have smooth or irregular contours⁵⁴.

Rectovaginal septum

Some studies have defined the TVS diagnosis of DIE in the RVS as absence of the normal appearance of the hyperechoic layer between the vagina and rectum due to the presence of a DIE nodule⁵⁵. Other researchers have used the terms 'RVS DIE' and 'rectovaginal DIE (RV DIE)' interchangeably to describe DIE in the RVS^{55,56}. The RVS is an individual anatomical structure with a specific location, whereas RV DIE describes DIE located in the rectovaginal area. The rectovaginal area includes the vagina, the rectum and the RVS. Furthermore, there is inconsistency in the definition of RV DIE in the literature. RV DIE has been described as endometriotic lesions which infiltrate both the rectum and the posterior vaginal fornix with possible extension into the RVS⁵⁵. Others have used the term 'rectovaginal endometriosis' to describe nodules which primarily infiltrate the RVS with possible extension into the vagina and/or rectum. Isolated RVS endometriosis is uncommon.

We propose that involvement of the RVS should be suspected when a DIE nodule is seen on TVS in the rectovaginal space below the line passing along the lower border of the posterior lip of the cervix (under the peritoneum)³⁹ (Figure 4). Isolated RVS DIE is rare (Figure 5); RVS DIE is usually an extension of posterior vaginal wall (Figure 6), anterior rectal wall (Figure 7) or both posterior vaginal wall and anterior rectal wall involvement⁵⁷ (Figure 8). The use of sonovaginography improves the detection of posterior vaginal and RVS

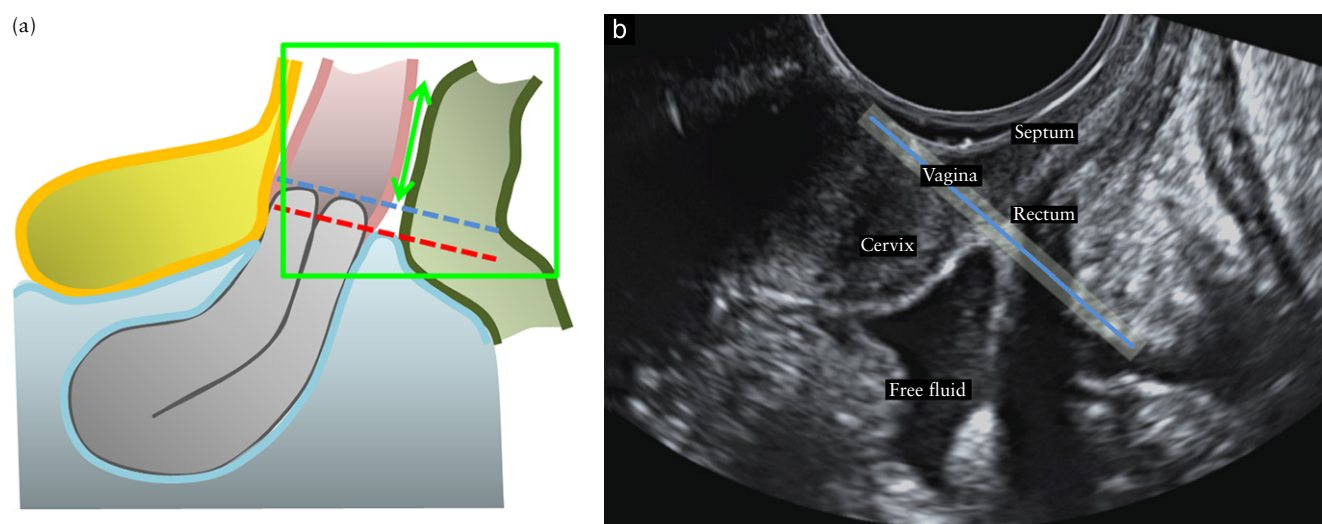


Figure 4 Schematic drawing (a) and ultrasound image (b) demonstrating our proposed ultrasound definition of the rectovaginal septum (RVS). (a) The RVS is denoted by the double-headed green arrow, below (anatomically) the blue line passing along the lower border of the posterior lip of the cervix. The posterior vaginal fornix lies between the blue line and the red line (the latter passing along the caudal end of the peritoneum of the lower margin of the rectouterine peritoneal pouch (cul-de-sac of Douglas)). (b) The upper delimitation of the RVS is where the blue line passes along the lower border of the posterior lip of the cervix.

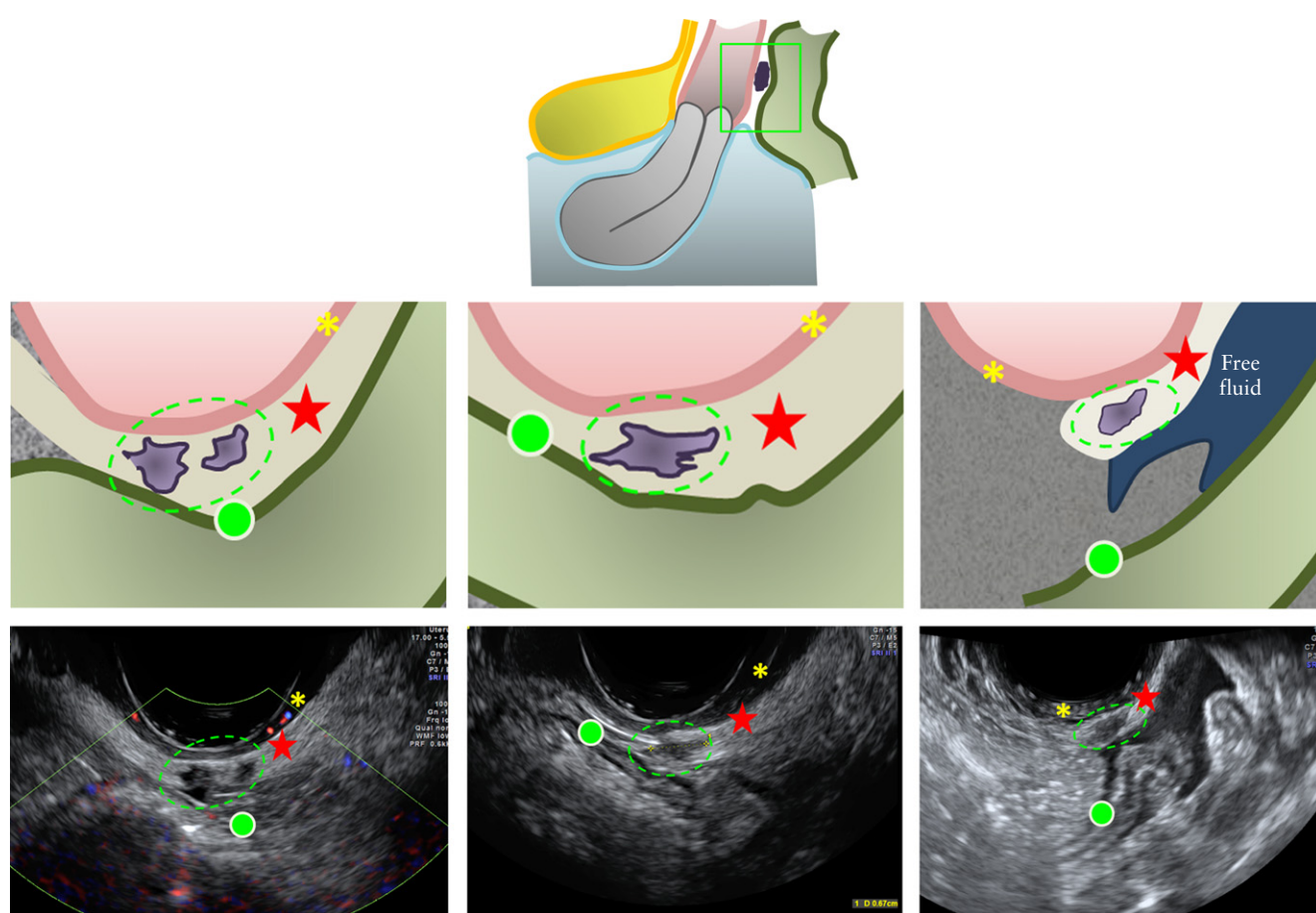


Figure 5 Schematic drawings and ultrasound images demonstrating isolated deep infiltrating endometriosis in the rectovaginal septum (RVS, ★). The green ellipses encircle the endometriotic nodules in the RVS. ●, bowel wall; ★, vaginal wall.

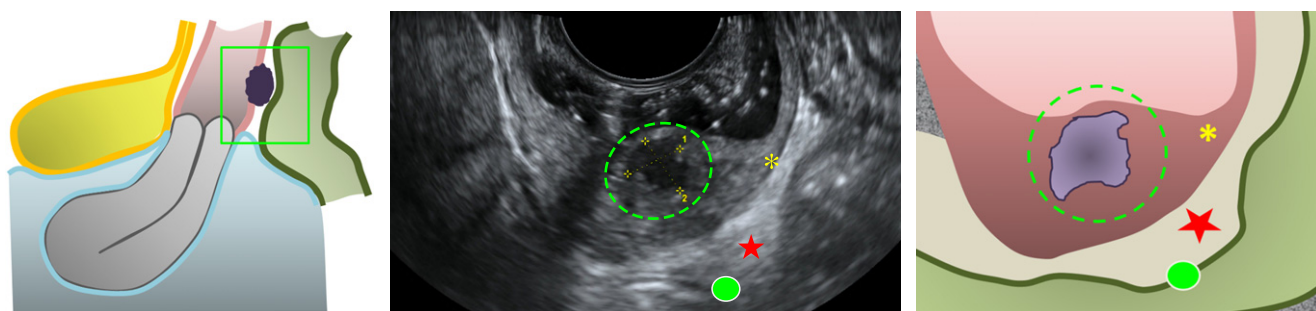


Figure 6 Schematic drawings and ultrasound image demonstrating deep infiltrating endometriosis in the posterior vaginal wall with extension into the rectovaginal septum (RVS, ★). The green ellipses encircle the endometriotic nodules in the RVS. ●, bowel wall; *, vaginal wall.

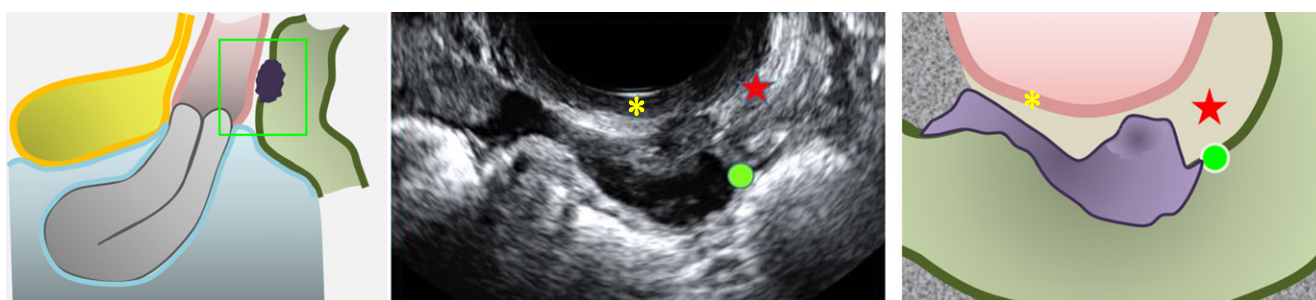


Figure 7 Schematic drawings and ultrasound image demonstrating deep infiltrating endometriosis in the anterior rectal wall with extension into the rectovaginal septum (★). ●, bowel wall; *, vaginal wall.

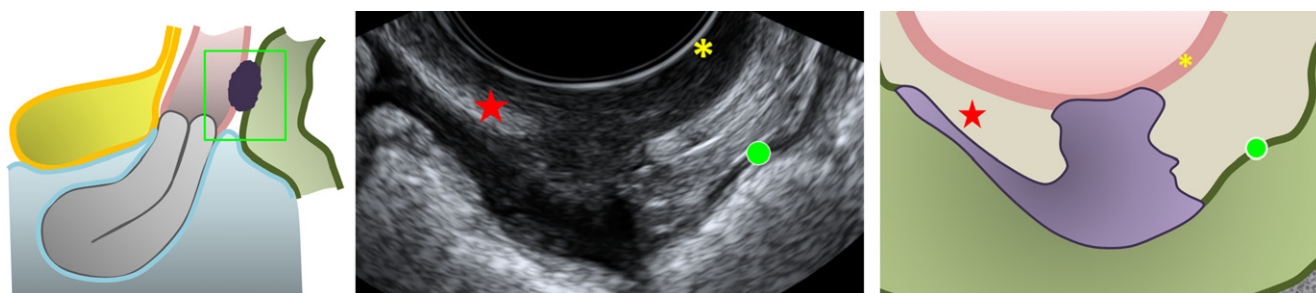


Figure 8 Schematic drawings and ultrasound image demonstrating rectovaginal septal (★) deep infiltrating endometriosis with extension into both anterior rectal wall and posterior vaginal wall. ●, bowel wall; *, vaginal wall.

DIE^{54,58}. The dimensions of the RVS DIE nodule should be recorded in three orthogonal planes and the distance between the lower margin of the lesion and the anal verge should be measured. This should be done whether the DIE is only in the vagina or only in the rectum, or involves the vagina, RVS and rectum. Low RVS lesions, when managed surgically, are associated with severe complications, including fistulae^{56,59–61}.

Vaginal wall

We propose that involvement of the posterior vaginal fornix and/or lateral vaginal fornix should be suspected when a DIE nodule is seen on TVS in the rectovaginal space below the line passing along the caudal end of the peritoneum of the lower margin of the rectouterine peritoneal pouch (cul-de-sac of Douglas) and above the

line passing along the lower border of the posterior lip of the cervix (under the peritoneum) (seen in Figure 4). Posterior vaginal fornix or forniceal endometriosis is suspected if the posterior vaginal fornix is thickened or if a discrete nodule is found in the hypoechoic layer of the vaginal wall (Figure S8a). The hypoechoic nodule may be homogeneous or inhomogeneous with or without large cystic areas (Figure S8a) and there may or may not be cystic areas surrounding the nodule^{6,39,41,42}. Figure S8b is an ultrasound image demonstrating posterior vaginal fornix DIE. The dimensions of the vaginal wall DIE nodule should be measured in three orthogonal planes.

Rectovaginal nodules ('diabolo'-like nodules)

Hourglass-shaped or 'diabolo'-like nodules occur when DIE lesions in the posterior vaginal fornix extend into

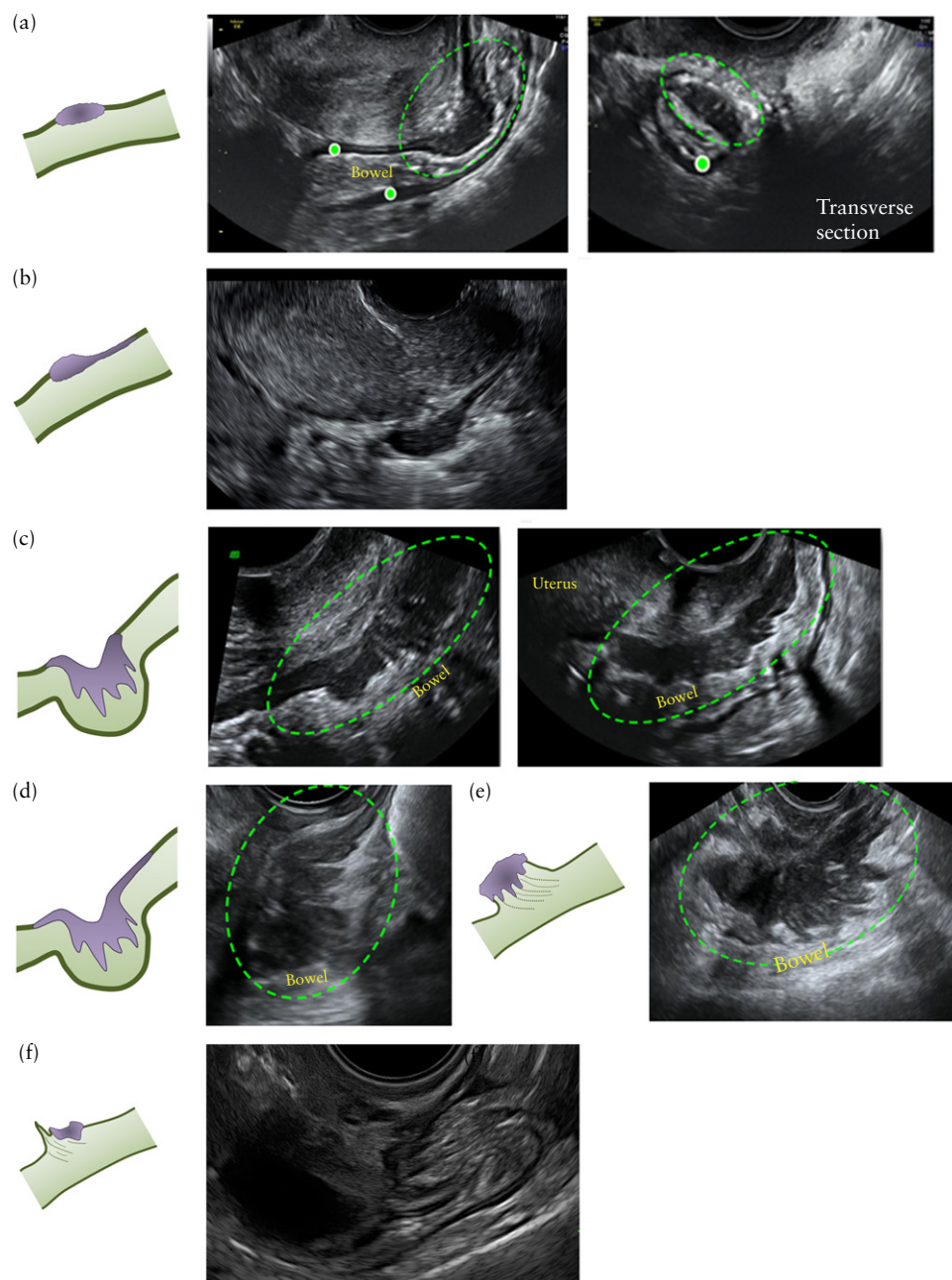


Figure 9 Schematic drawings and corresponding ultrasound images of bowel deep infiltrating endometriosis (DIE). (a) DIE nodule with a regular outline (absence of 'spikes'). (b) DIE nodule with progressive narrowing, like a 'tail', also known as 'comet' sign. (c) DIE nodule with prominent spikes towards the bowel lumen, also known as 'Indian headdress' or 'moose antler' sign. (d) DIE nodule with both prominent spikes towards the bowel lumen (Indian headdress/moose antler sign) and progressive narrowing like a tail (comet sign). (e) DIE nodule with both prominent spikes towards the bowel lumen (Indian headdress/moose antler sign) and extrinsic retraction (and visible mucosal folds) (known as 'pulling sleeve' sign). The sliding sign is expected to be negative. (f) DIE nodule and extrinsic retraction (pulling sleeve sign). The sliding sign is expected to be negative. Ultrasound image shows bowel adherent to the ovary; the hypoechogenic area between the bowel loops and the ovary to which the bowel is adherent correspond to linear endometriotic changes (containing endometriotic glands and stroma) not involving the muscularis of the bowel wall but located between the bowel and the ovary.

the anterior rectal wall⁶² (Figure S9a). On ultrasound, the part of the DIE lesion situated in the anterior rectal wall is the same size as the part situated in the posterior vaginal fornix (Figure S9b). There is a small but easily visualized continuum between these two parts of the lesion. These lesions are located below the peritoneum of the POD and are usually large (3 cm on average)⁶³.

Uterosacral ligaments

Normal USLs are usually not visible on ultrasound (Figure S10a). USL DIE lesions can be seen in the mid-sagittal view of the uterus (Figure S10b). However, these are best seen by placing the transvaginal probe in the posterior vaginal fornix in the midline in the sagittal plane and then sweeping the probe inferolaterally to the cervix. USLs

are considered to be affected by DIE when a hypoechoic thickening with regular or irregular margins is seen within the peritoneal fat surrounding the USLs. The lesion may be isolated or may be part of a larger nodule extending into the vagina or into other surrounding structures. The thickness of a 'thickened' USL can be measured in the transverse plane at the insertion of the ligament on the cervix provided that the ligament can be distinguished clearly from adjacent structures (Figure S10c). In some cases the DIE lesion involving the USL is located at the torus uterinus (Figure S10d). If so, it is seen as a central thickening of the retrocervical area⁶⁴. The dimensions of the USL DIE nodule should be recorded in three orthogonal planes.

Rectum, rectosigmoid junction and sigmoid

Bowel DIE classically involves the anterior rectum, rectosigmoid junction and/or sigmoid colon, all of which can be visualized using TVS. Figure S11a demonstrates a schematic drawing of a DIE lesion within the upper anterior rectum. Bowel DIE can take the form of an isolated lesion or can be multifocal (multiple lesions affecting the same segment) and/or multicentric (multiple lesions affecting several bowel segments, i.e. small bowel, large bowel, cecum, ileocecal junction and/or appendix)⁶⁵. Although TVS can be used to visualize multifocal rectal DIE (Figure S11b), there are no published data assessing its performance. Computed tomographic colonography and magnetic resonance imaging (MRI) can be used to diagnose both multifocal and multicentric bowel endometriosis⁶⁵.

Histologically, bowel endometriosis is defined as the presence of endometrial glands and stroma in the bowel wall, reaching at least the muscularis propria⁶⁶, where this invariably induces smooth-muscle hyperplasia and fibrosis. This results in thickening of the bowel wall and some narrowing of the bowel lumen. Normal rectal wall layers can be visualized on TVS: the anterior rectal serosa is seen as a thin hyperechoic line; the muscularis propria is hypoechoic, with the longitudinal smooth muscle (outer) and circular smooth muscle (inner) separated by a faint thin hyperechoic line; the submucosa is hyperechogenic; and the mucosa is hypoechoic^{37,67} (Figure S12a). Bowel DIE usually appears on TVS as a thickening of the hypoechoic muscularis propria or as hypoechoic nodules, with or without hyperechoic foci (Figure S12b) with blurred margins. The morphological type of bowel lesion should be described according to Figure 9. Sonographically, bowel lesions are hypoechoic and in some cases a thinner section or a 'tail' is noted at one end, resembling a 'comet'⁶⁸ (Figure 9b). The normal appearance of the muscularis propria of the rectum or rectosigmoid is replaced by a nodule of abnormal tissue with possible retraction and adhesions, resulting in the so-called 'Indian headdress' or 'moose antler' sign (Figure 9c,e,f)⁴²; the size of these lesions can vary.

We propose that bowel DIE lesions noted on TVS be described according to the segment of the rectum or sigmoid colon in which they occur, with DIE lesions

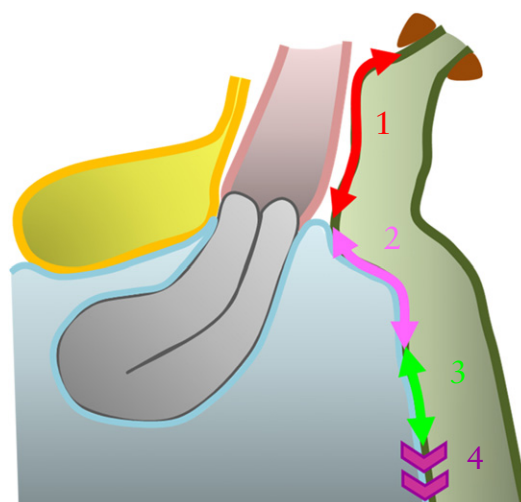


Figure 10 Schematic drawing demonstrating distinction at ultrasound between segments of the rectum and sigmoid colon for specifying location of deep infiltrating endometriotic lesions: lower (or retroperitoneal) anterior rectum (1); upper (visible at laparoscopy) anterior rectum (2); rectosigmoid junction (3); and anterior sigmoid (4).

located below the level of the insertion of the USLs on the cervix being denoted as lower (retroperitoneal) anterior rectal DIE lesions, those above this level being denoted as upper (visible at laparoscopy) anterior rectal DIE lesions, those at the level of the uterine fundus being denoted as rectosigmoid junction DIE lesions and those above the level of the uterine fundus being denoted as anterior sigmoid DIE lesions (Figure 10). The dimensions of the rectal and/or rectosigmoid DIE nodules should be recorded in three orthogonal planes and the distance between the lower margin of the most caudal lesion and the anal verge should be measured using TVS.

Because bowel DIE may affect the bowel simultaneously at different sites, other bowel lesions should be looked for carefully when there is a DIE lesion affecting the rectum (Figure S12b) or rectosigmoid. Preliminary data showed that rectal DIE lesions may be associated with a second intestinal lesion in 54.6% of cases³⁴.

Ultrasound diagnosis of POD obliteration^{31,32} has been explained extensively earlier in this article. The obliteration can be graded as partial or complete depending on whether one side (left or right) or both sides, respectively, demonstrate a negative sliding sign. Furthermore, an experienced operator can identify the level of POD obliteration, i.e. specifying, in an anteverted uterus, whether it is at the retrocervical level (lower third of the uterus), mid-posterior uterus (middle third) and/or posterior uterine fundus (upper third)⁶⁹ and, in a retroverted uterus, whether it is at the posterior uterine fundus, mid-anterior uterus and/or lower anterior uterine wall³³ (Figure S13).

MEASUREMENT OF LESIONS

We propose that each endometrioma and DIE lesion should be measured systematically in three orthogonal

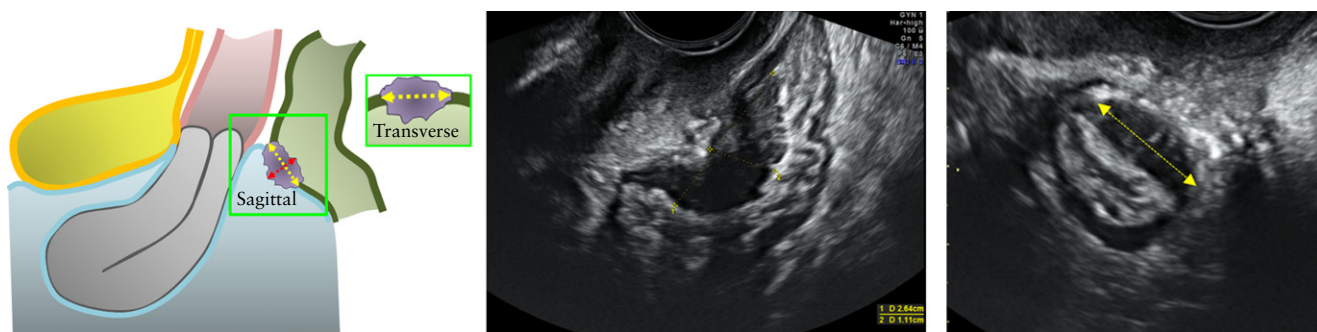


Figure 11 Schematic drawing and ultrasound images demonstrating measurement of a nodule of deep infiltrating endometriosis in the bowel wall. Three orthogonal measurements should be taken, i.e. mid-sagittal, anteroposterior and transverse.

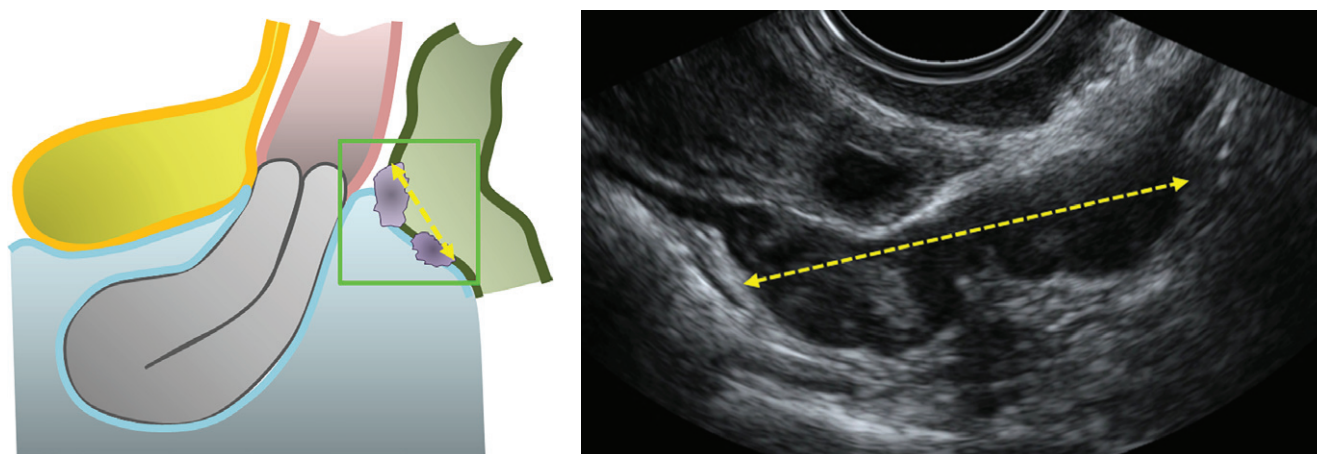


Figure 12 In cases of multifocal lesions of deep infiltrating endometriosis in the bowel, the total length of the bowel segment involved (from caudal to cephalic aspect) should be measured, as shown in this schematic drawing and ultrasound image.

planes, to obtain the length (mid-sagittal measurement), thickness (anteroposterior measurement) and transverse diameter (Figure 11). This approach of measuring in three planes applies to DIE lesions located in the bladder, RVS, vagina, USLs, anterior rectum and rectosigmoid.

Additionally, in cases of endometriosis in the ureters, it is important to measure the distance between the distal ureteric orifice and a DIE lesion which causes a ureteric stricture; the stricture can be caused by either extrinsic compression or intrinsic infiltration. Once the stricture is identified along the longitudinal course of the ureter, one caliper should be placed at this level and the other at the distal ureteric orifice for measurement (Figure S7).

In cases of multifocal bowel DIE lesions the total mid-sagittal length of the bowel segment involved, from caudal to cephalic aspect, should be measured (Figure 12). It is important to be aware that the retraction within rectosigmoid DIE lesions can result in an overestimation of the true thickness of the lesion and an underestimation of the true length of the lesion (Figure S14). This has been described as the 'mushroom cap' sign on MRI and can also be noted on TVS⁷⁰.

In cases of DIE lesions in the bowel or RVS, it is important to measure the distance between the anal verge

and the lesion (Figure S15). It is possible to measure the distance from the anus to the bowel lesion using transrectal sonography. By inserting the probe into the anus and positioning the tip of the probe up against the endometriotic lesion⁷¹, one's finger can be kept on the probe at the level of the anus and a ruler used to measure the distance from the finger on the probe to the tip of the probe when the probe has been withdrawn. TVS can also be used to approximate the distance from the anal verge to the lower margin of the bowel lesion. If there are multifocal bowel lesions, then the distance between the anal verge and the most caudal bowel lesion is measured.

Figure 13 gives an overview of anterior and posterior compartmental locations for deep infiltrating endometriosis.

OTHER ULTRASOUND TECHNIQUES

Color Doppler

Although well established in the evaluation of endometrioma²⁰, no prospective data have been reported for the role of color Doppler in the evaluation of DIE. Usually, endometriotic lesions in the rectosigmoid are poorly vascularized. Color Doppler is useful in the differential diagnosis between DIE in the bowel and rectal

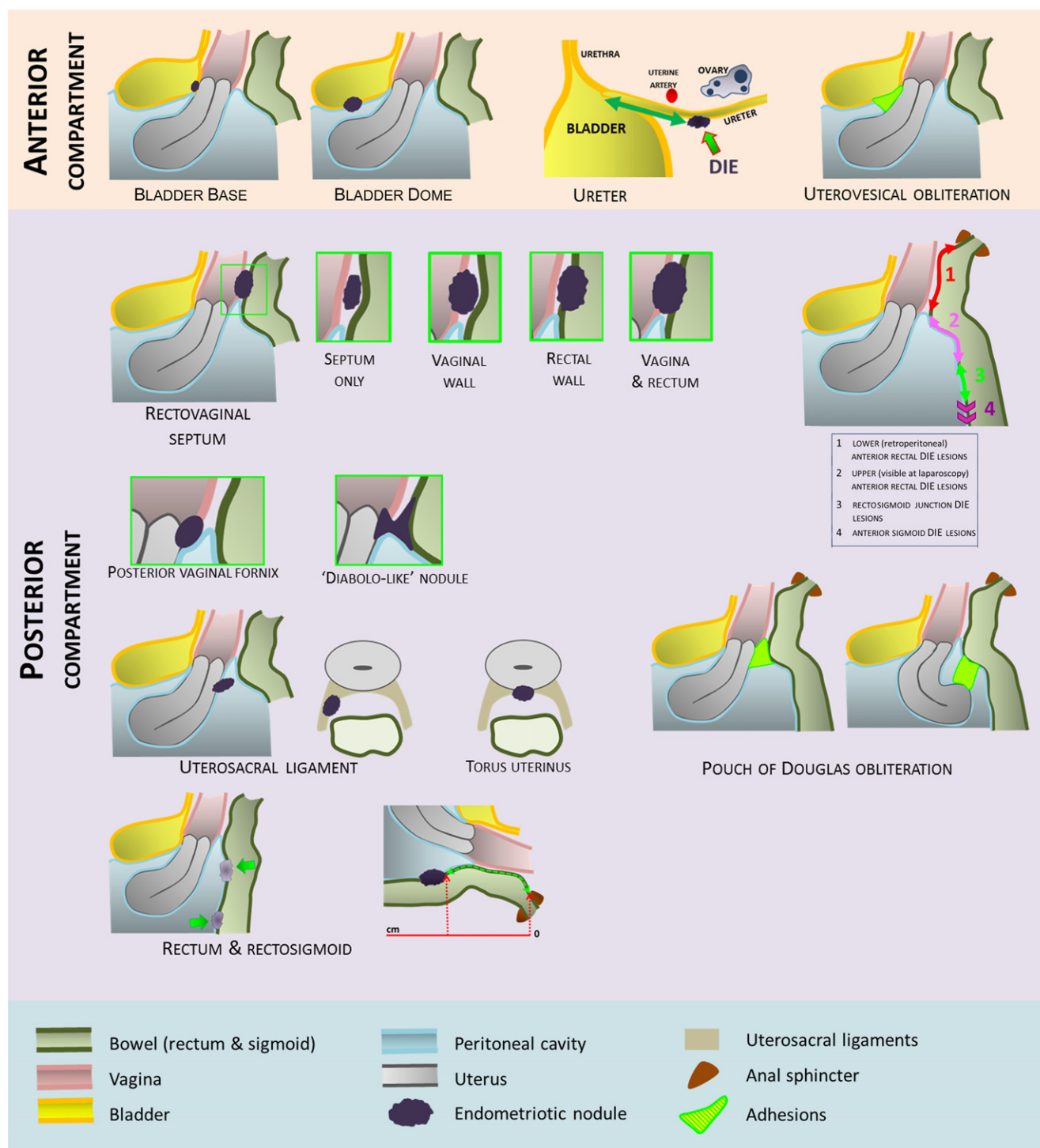


Figure 13 Schematic drawings giving overview of anterior and posterior compartmental locations of deep infiltrating endometriosis.

cancer (Figure S16) and we propose that color Doppler be used as an adjunct modality in the assessment of DIE lesions of the bowel.

Tenderness-guided ultrasound examination

Tenderness-guided ultrasound examination is performed with or without an acoustic window between the transvaginal probe and the surrounding vaginal structures, coupled with an 'active' role of the patient, who

indicates the site of any tenderness experienced during the examination^{27,41}.

Rectal water contrast transvaginal ultrasound

Rectal water contrast TVS requires injection, under ultrasound guidance, of saline through a catheter into the rectum; the procedure is well tolerated^{36,72} and allows estimation of the degree of stenosis of the bowel lumen⁷³ (Figure S17).

Sonovaginography with saline or gel

Saline contrast sonovaginography combines TVS with injection of saline into the vagina⁷⁴. A transvaginal probe is used with, at its base, a purpose-designed hydraulic ring that inflates with approximately 40 mL saline solution to prevent escape of the 60–120 mL saline that is injected into the vagina using a Foley catheter⁵⁸. The solution creates an acoustic window between the transvaginal probe and the structures surrounding the vagina and exerts pressure that distends the vaginal walls. This permits more complete visualization of the vaginal walls and anterior/posterior vaginal fornices.

In order to perform gel contrast sonovaginography, 20–50 mL ultrasound gel is inserted into the posterior vaginal fornix, using a 20-mL plastic syringe, before insertion of the transvaginal probe^{35,57}. The gel creates an acoustic window, allowing a ‘stand-off’ view of the structures of the posterior compartment (Figure S18). The gel must be loaded carefully into the syringe, ensuring there are no or only minimal air bubbles in the gel. The syringe is filled completely, so that the plunger comes in direct contact with the gel, reducing the possibility of air pockets when instilling the gel into the vagina. Great care is taken to ensure that the syringe is inserted sufficiently far into the vagina that the gel fills the posterior fornix completely. In published reports, no woman required any refilling of the posterior fornix with gel^{35,57}.

Transrectal sonography using transvaginal probe

Transrectal sonography^{61,75} can be used if necessary; if TVS is impossible or inappropriate, for example if the woman is *virgo intacta*.

Three-dimensional ultrasound

Three-dimensional TVS

In one study, offline analysis of three-dimensional (3D) volumetric datasets was useful in the diagnosis of posterior locations of DIE without intestinal involvement, such as DIE in the USLs, vagina or RVS¹⁰. 3D ultrasound, however, does not permit evaluation of the mobility of pelvic organs; nor does it allow mapping of SST.

Introital three-dimensional sonography

There are no studies demonstrating that 3D introital ultrasound outperforms 2D ultrasound in the detection or characterization of DIE. However, one research team reported introital 3D sonography to be an effective and reproducible method for detecting and describing endometriosis in the RVS^{76,77} (Figure S19). Guerriero *et al.*⁷⁸ suggested that 3D image rendering allows analysis of DIE nodules because 3D rendering may show irregular shapes and borders clearly.

Three-dimensional rectosonography

There are no studies demonstrating that 3D rectosonography outperforms 2D ultrasound in the detection or

characterization of DIE. However, in a recent study, 3D rectosonography was found to be highly concordant with MRI⁷⁹.

Transvaginal elastography

Limited data are available on the usefulness of transvaginal elastography in the diagnosis of DIE⁸⁰. DIE nodules usually demonstrate high stiffness on elastography (Figure S20).

CONCLUSION

TVS is the first-line investigative tool in the work-up of women with potential underlying endometriosis³. The ability for ultrasound to detect ovarian endometriosis and DIE (bowel and non-bowel) is well documented^{4,5,8,19}. Prediction of severe forms of DIE as well as POD obliteration using TVS is helpful in planning a multidisciplinary surgical approach^{3,16,22,27,30,31,56,81}.

Some experience with gynecological ultrasound is necessary to assess the sliding sign to predict POD obliteration. Menakaya *et al.*⁸² found that trainees in obstetrics and gynecology who have performed at least 200 prior TVS scans outperform trainees who have performed fewer than 200 in interpreting offline videos of the sliding sign. They also found that interpretation of the sliding sign at the retrocervix was better than that at the posterior upper uterine fundus. Experienced operators who have performed in excess of 2500 scans reach proficiency in performing the sliding sign maneuver and detecting POD obliteration after approximately 40 examinations^{83,84}.

Inter- and intraobserver agreement and diagnostic accuracy with regard to interpretation of the TVS sliding sign to predict POD obliteration has been found to be acceptable, with agreement ranging from substantial to almost perfect for observers who specialize in gynecological ultrasound⁸⁵. In the same study, the agreement for all observers was higher for interpretation of the sliding sign in the retrocervical region compared with the posterior upper uterine fundal region.

Similar to detection of POD obliteration, experienced operators who have performed in excess of 2500 scans reach proficiency in the detection of rectal DIE nodules using TVS after approximately 40 examinations^{83,84}. With the exception of DIE affecting the RVS, TVS in the hands of well-trained staff is a highly accurate and reproducible method for non-invasive diagnosis of DIE⁸⁶.

In this consensus opinion, we have described a systematic approach to examining the pelvis in women with suspected endometriosis, and defined terms and measurements to describe the appearance of endometriosis on sonography. This consensus opinion represents the collective opinion of clinicians, gynecological sonologists, advanced laparoscopic surgeons and radiologists with an interest in diagnosis and management of endometriosis. Currently, it is difficult to compare results between published studies, because authors use different terms when describing the same structures and locations.

We hope that the terms and definitions suggested herein will be adopted in centers around the world. This would result in consistent use of nomenclature when describing the ultrasound location and extent of endometriosis. We believe that the standardization of terminology should allow meaningful comparisons between future studies in women with an ultrasound diagnosis of endometriosis and should facilitate multicenter studies.

REFERENCES

- Ferrero S, Arena E, Morando A, Remorgida V. Prevalence of newly diagnosed endometriosis in women attending the general practitioner. *Int J Gynaecol Obstet* 2010; **110**: 203–207.
- Johnson NP, Hummelshoj L, World Endometriosis Society Montpellier C. Consensus on current management of endometriosis. *Hum Reprod* 2013; **28**: 1552–1568.
- Piketty M, Chopin N, Dousset B, Millischer-Bellaische AE, Roseau G, Leconte M, Borghese B, Chapron C. Preoperative work-up for patients with deeply infiltrating endometriosis: transvaginal ultrasonography must definitely be the first-line imaging examination. *Hum Reprod* 2009; **24**: 602–607.
- Guerriero S, Ajossa S, Minguez JA, Jurado M, Mais V, Melis GB, Alcazar JL. Accuracy of transvaginal ultrasound for diagnosis of deep endometriosis in uterosacral ligaments, rectovaginal septum, vagina and bladder: systematic review and meta-analysis. *Ultrasound Obstet Gynecol* 2015; **46**: 534–45.
- Guerriero S, Ajossa S, Orozco R, Perniciano M, Jurado M, Melis GB, Alcazar JL. Accuracy of transvaginal ultrasound for diagnosis of deep endometriosis in the rectosigmoid: systematic review and meta-analysis. *Ultrasound Obstet Gynecol* 2016; **47**: 281–289.
- Hudelist G, Ballard K, English J, Wright J, Banerjee S, Mastoroudes H, Thomas A, Singer CF, Keckstein J. Transvaginal sonography vs clinical examination in the preoperative diagnosis of deep infiltrating endometriosis. *Ultrasound Obstet Gynecol* 2011; **37**: 480–487.
- Hudelist G, Oberwinkler KH, Singer CF, Tuttlies F, Rauter G, Ritter O, Keckstein J. Combination of transvaginal sonography and clinical examination for preoperative diagnosis of pelvic endometriosis. *Hum Reprod* 2009; **24**: 1018–1024.
- Hudelist G, English J, Thomas AE, Tinelli A, Singer CF, Keckstein J. Diagnostic accuracy of transvaginal ultrasound for non-invasive diagnosis of bowel endometriosis: systematic review and meta-analysis. *Ultrasound Obstet Gynecol* 2011; **37**: 257–263.
- Rossi L, Palazzo L, Yazbeck C, Walker F, Chis C, Luton D, Koskas M. Can rectal endoscopic sonography be used to predict infiltration depth in patients with deep infiltrating endometriosis of the rectum? *Ultrasound Obstet Gynecol* 2014; **43**: 322–327.
- Guerriero S, Saba L, Ajossa S, Peddes C, Angiolucci M, Perniciano M, Melis GB, Alcazar JL. Three-dimensional ultrasonography in the diagnosis of deep endometriosis. *Hum Reprod* 2014; **29**: 1189–1198.
- Savelli L, Manuzzi L, Coe M, Mabrouk M, Di Donato N, Venturoli S, Seracchioli R. Comparison of transvaginal sonography and double-contrast barium enema for diagnosing deep infiltrating endometriosis of the posterior compartment. *Ultrasound Obstet Gynecol* 2011; **38**: 466–471.
- Guerriero S, Abreo M, Hudelist G, Bazot M, Ajossa S, Alcazar J, Pascual M, Condous G, Ferrero S, Dessole S, Leone F, Timmerman D. Ultrasonography in deep endometriosis: a consensus opinion from the International Deep Endometriosis Analysis (IDEA) group. A preliminary statement. *Ultrasound Obstet Gynecol* 2011; **38**: 265.
- Chapron C, Fauconnier A, Vieira M, Barakat H, Dousset B, Pansini V, Vacher-Lavenu MC, Dubuisson JB. Anatomical distribution of deeply infiltrating endometriosis: surgical implications and proposition for a classification. *Hum Reprod* 2003; **18**: 157–161.
- Chapron C, Barakat H, Fritel X, Dubuisson JB, Breart G, Fauconnier A. Presurgical diagnosis of posterior deep infiltrating endometriosis based on a standardized questionnaire. *Hum Reprod* 2005; **20**: 507–513.
- Fedele L, Bianchi S, Carmignani L, Berlanda N, Fontana E, Frontino G. Evaluation of a new questionnaire for the presurgical diagnosis of bladder endometriosis. *Hum Reprod* 2007; **22**: 2698–2701.
- Menakaya U, Reid S, Infante F, Condous G. Systematic evaluation of women with suspected endometriosis using a 5-domain sonographically based approach. *J Ultrasound Med* 2015; **34**: 937–947.
- Holland TK, Cutner A, Saridogan E, Mavrellos D, Pateman K, Jurkovic D. Ultrasound mapping of pelvic endometriosis: does the location and number of lesions affect the diagnostic accuracy? A multicentre diagnostic accuracy study. *BMC Womens Health* 2013; **13**: 43.
- Di Donato N, Bertoldo V, Montanari G, Zannoni L, Caprara G, Seracchioli R. Question mark form of uterus: a simple sonographic sign associated with the presence of adenomyosis. *Ultrasound Obstet Gynecol* 2015; **46**: 126–127.
- Van den Bosch T, Dueholm M, Leone FP, Valentin L, Rasmussen CK, Votino A, Van Schoubroeck D, Landolfo C, Installe AJ, Guerriero S, Exacoustos C, Gordts S, Benacerraf B, D'Hooghe T, De Moor B, Brolmann H, Goldstein S, Epstein E, Bourne T, Timmerman D. Terms, definitions and measurements to describe sonographic features of myometrium and uterine masses: a consensus opinion from the Morphological Uterus Sonographic Assessment (MUSA) group. *Ultrasound Obstet Gynecol* 2015; **46**: 284–298.
- Van Holsbeke C, Van Calster B, Guerriero S, Savelli L, Paladini D, Lissoni AA, Czekierdowski A, Fischerova D, Zhang J, Mestdagh G, Testa AC, Bourne T, Valentin L, Timmerman D. Endometriomas: their ultrasound characteristics. *Ultrasound Obstet Gynecol* 2010; **35**: 730–740.
- Timmerman D, Valentin L, Bourne TH, Collins WP, Verralst H, Vergote I, International Ovarian Tumor Analysis G. Terms, definitions and measurements to describe the sonographic features of adnexal tumors: a consensus opinion from the International Ovarian Tumor Analysis (IOTA) Group. *Ultrasound Obstet Gynecol* 2000; **16**: 500–505.
- Redwine DB. Ovarian endometriosis: a marker for more extensive pelvic and intestinal disease. *Fertil Steril* 1999; **72**: 310–315.
- Chapron C, Pietin-Vialle C, Borghese B, Davy C, Foulot H, Chopin N. Associated ovarian endometrioma is a marker for greater severity of deeply infiltrating endometriosis. *Fertil Steril* 2009; **92**: 453–457.
- Ghezzi F, Raio L, Cromi A, Duwe DG, Beretta P, Buttarelli M, Mueller MD. “Kissing ovaries”: a sonographic sign of moderate to severe endometriosis. *Fertil Steril* 2005; **83**: 143–147.
- Masilini F, Moruzzi C, Giansiracusa C, Guastafierro F, Savelli L, De Meis L, Epstein E, Timor-Tritsch IE, Mailath-Pokorny M, Ercoli A, Exacoustos C, Benacerraf BR, Valentin L, Testa AC. Imaging in gynecological disease. 10: Clinical and ultrasound characteristics of decidualized endometriomas surgically removed during pregnancy. *Ultrasound Obstet Gynecol* 2014; **44**: 354–360.
- Guerriero S, Ajossa S, Lai MP, Mais V, Paoletti AM, Melis GB. Transvaginal ultrasonography in the diagnosis of pelvic adhesions. *Hum Reprod* 1997; **12**: 2649–2653.
- Okaro E, Condous G, Khalid A, Timmerman D, Ameye L, Huffel SV, Bourne T. The use of ultrasound-based ‘soft markers’ for the prediction of pelvic pathology in women with chronic pelvic pain—can we reduce the need for laparoscopy? *BJOG* 2006; **113**: 251–256.
- Exacoustos C, Zupi E, Carusotti C, Rinaldo D, Marconi D, Lanzi G, Arduini D. Staging of pelvic endometriosis: role of sonographic appearance in determining extension of disease and modulating surgical approach. *J Am Assoc Gynecol Laparosc* 2003; **10**: 378–382.
- Guerriero S, Ajossa S, Garau N, Alcazar JL, Mais V, Melis GB. Diagnosis of pelvic adhesions in patients with endometrioma: the role of transvaginal ultrasonography. *Fertil Steril* 2010; **94**: 742–746.
- Holland TK, Yazbek J, Cutner A, Saridogan E, Hoo WL, Jurkovic D. Value of transvaginal ultrasound in assessing severity of pelvic endometriosis. *Ultrasound Obstet Gynecol* 2010; **36**: 241–248.
- Hudelist G, Fritzer N, Staettner S, Tammaa A, Tinelli A, Sparic R, Keckstein J. Uterine sliding sign: a simple sonographic predictor for presence of deep infiltrating endometriosis of the rectum. *Ultrasound Obstet Gynecol* 2013; **41**: 692–695.
- Reid S, Lu C, Casikar I, Reid G, Abbott J, Cario G, Chou D, Kowalski D, Cooper M, Condous G. Prediction of pouch of Douglas obliteration in women with suspected endometriosis using a new real-time dynamic transvaginal ultrasound technique: the sliding sign. *Ultrasound Obstet Gynecol* 2013; **41**: 685–691.
- Menakaya U, Condous G. The retroverted uterus: refining the description of the real time dynamic ‘sliding sign’. *Aust J Ultrasound Med* 2013; **16**: 97.
- Goncalves MO, Podgaec S, Dias JA, Jr, Gonzalez M, Abreo MS. Transvaginal ultrasonography with bowel preparation is able to predict the number of lesions and rectosigmoid layers affected in cases of deep endometriosis, defining surgical strategy. *Hum Reprod* 2010; **25**: 665–671.
- Leon M, Vaccaro H, Alcazar JL, Martinez J, Gutierrez J, Amor F, Iturra A, Sovino H. Extended transvaginal sonography in deep infiltrating endometriosis: use of bowel preparation and an acoustic window with intravaginal gel: preliminary results. *J Ultrasound Med* 2014; **33**: 315–321.
- Valenzano Menada M, Remorgida V, Abbamonte LH, Nicoletti A, Ragni N, Ferrero S. Does transvaginal ultrasonography combined with water-contrast in the rectum aid in the diagnosis of rectovaginal endometriosis infiltrating the bowel? *Hum Reprod* 2008; **23**: 1069–1075.
- Chamie LP, Pereira RM, Zanatta A, Serafini PC. Transvaginal US after bowel preparation for deeply infiltrating endometriosis: protocol, imaging appearances, and laparoscopic correlation. *Radiographics* 2010; **30**: 1235–1249.
- Savelli L, Manuzzi L, Pollastri P, Mabrouk M, Seracchioli R, Venturoli S. Diagnostic accuracy and potential limitations of transvaginal sonography for bladder endometriosis. *Ultrasound Obstet Gynecol* 2009; **34**: 595–600.
- Bazot M, Thomassin I, Hourani R, Cortez A, Darai E. Diagnostic accuracy of transvaginal sonography for deep pelvic endometriosis. *Ultrasound Obstet Gynecol* 2004; **24**: 180–185.
- Abreo MS, Goncalves MO, Dias JA, Jr, Podgaec S, Chamie LP, Blasbalg R. Comparison between clinical examination, transvaginal sonography and magnetic resonance imaging for the diagnosis of deep endometriosis. *Hum Reprod* 2007; **22**: 3092–3097.
- Guerriero S, Ajossa S, Gerada M, D'Aquila M, Piras B, Melis GB. “Tenderness-guided” transvaginal ultrasonography: a new method for the detection of deep endometriosis in patients with chronic pelvic pain. *Fertil Steril* 2007; **88**: 1293–1297.
- Guerriero S, Ajossa S, Gerada M, Virgilio B, Angioni S, Melis GB. Diagnostic value of transvaginal ‘tenderness-guided’ ultrasonography for the prediction of location of deep endometriosis. *Hum Reprod* 2008; **23**: 2452–2457.
- Fedele L, Bianchi S, Raffaelli R, Portuese A. Pre-operative assessment of bladder endometriosis. *Hum Reprod* 1997; **12**: 2519–2522.
- Moro F, Mavrellos D, Pateman K, Holland T, Hoo WL, Jurkovic D. Prevalence of pelvic adhesions on ultrasound examination in women with a history of Cesarean section. *Ultrasound Obstet Gynecol* 2015; **45**: 223–228.
- Pateman K, Mavrellos D, Hoo WL, Holland T, Naftalin J, Jurkovic D. Visualization of ureters on standard gynecological transvaginal scan: a feasibility study. *Ultrasound Obstet Gynecol* 2013; **41**: 696–701.
- Pateman K, Holland TK, Knez J, Derdelis G, Cutner A, Saridogan E, Jurkovic D. Should a detailed ultrasound examination of the complete urinary tract be routinely

- performed in women with suspected pelvic endometriosis? *Hum Reprod* 2015; 30: 2802–2807.
47. Knabben L, Imboden S, Fellmann B, Nirgianakis K, Kuhn A, Mueller MD. Urinary tract endometriosis in patients with deep infiltrating endometriosis: prevalence, symptoms, management, and proposal for a new clinical classification. *Fertil Steril* 2015; 103: 147–152.
 48. Carmignani L, Vercellini P, Spinelli M, Fontana E, Frontino G, Fedele L. Pelvic endometriosis and hydronephrosis. *Fertil Steril* 2010; 93: 1741–1744.
 49. Webb JA. Ultrasonography and Doppler studies in the diagnosis of renal obstruction. *BJU Int* 2000; 86 (Suppl 1): 25–32.
 50. Webb JA. Ultrasonography in the diagnosis of renal obstruction. *BMJ* 1990; 301: 944–946.
 51. Reid S, Condous G. Should ureteric assessment be included in the transvaginal ultrasound assessment for women with suspected endometriosis? *Aust J Ultrasound Med* 2015; 18: 2.
 52. Block B. *The Practice of Ultrasound: a Step-by-Step Guide to Abdominal Scanning*. Stuttgart, Germany: Thieme, 2011.
 53. Chapron C, Chopin N, Borghese B, Foulot H, Dousset B, Vacher-Lavenu MC, Vieira M, Hasan W, Bricou A. Deeply infiltrating endometriosis: pathogenetic implications of the anatomical distribution. *Hum Reprod* 2006; 21: 1839–1845.
 54. Dessole S, Farina M, Rubattu G, Cosmi E, Ambrosini G, Nardelli GB. Sonovaginography is a new technique for assessing rectovaginal endometriosis. *Fertil Steril* 2003; 79: 1023–1027.
 55. Reid S, Condous G. The issues surrounding the pre-operative TVS diagnosis of rectovaginal septum endometriosis. *Aust J Ultrasound Med* 2014; 17: 2–3.
 56. Martin DC, Batt RE. Retrocervical, retrovaginal pouch, and rectovaginal septum endometriosis. *J Am Assoc Gynecol Laparosc* 2001; 8: 12–17.
 57. Reid S, Lu C, Hardy N, Casikar I, Reid G, Cario G, Chou D, Almashat D, Condous G. Office gel sonovaginography for the prediction of posterior deep infiltrating endometriosis: a multicenter prospective observational study. *Ultrasound Obstet Gynecol* 2014; 44: 710–718.
 58. Saccardi C, Cosmi E, Borghero A, Tregnaghi A, Dessole S, Litta P. Comparison between transvaginal sonography, saline contrast sonovaginography and magnetic resonance imaging in the diagnosis of posterior deep infiltrating endometriosis. *Ultrasound Obstet Gynecol* 2012; 40: 464–469.
 59. Vercellini P, Parazzini F, Pietropaolo G, Cipriani S, Frattaruolo MP, Fedele L. Pregnancy outcome in women with peritoneal, ovarian and rectovaginal endometriosis: a retrospective cohort study. *BJOG* 2012; 119: 1538–1543.
 60. Donnez J, Nisolle M, Smoes P, Gillet N, Beguin S, Casanas-Roux F. Peritoneal endometriosis and “endometriotic” nodules of the rectovaginal septum are two different entities. *Fertil Steril* 1996; 66: 362–368.
 61. Fedele L, Bianchi S, Portuese A, Borrufo F, Dorta M. Transrectal ultrasonography in the assessment of rectovaginal endometriosis. *Obstet Gynecol* 1998; 91: 444–448.
 62. Squifflet J, Feger C, Donnez J. Diagnosis and imaging of adenomyotic disease of the retroperitoneal space. *Gynecol Obstet Invest* 2002; 54 (Suppl 1): 43–51.
 63. Donnez J, Pirard C, Smets M, Jadoul P, Squifflet J. Surgical management of endometriosis. *Best Pract Res Clin Obstet Gynaecol* 2004; 18: 329–348.
 64. Bazot M, Darai E, Hourani R, Thomassin I, Cortez A, Uzan S, Buy JN. Deep pelvic endometriosis: MR imaging for diagnosis and prediction of extension of disease. *Radiology* 2004; 232: 379–389.
 65. Belghiti J, Thomassin-Naggara I, Zacharopoulou C, Zilberman S, Jarboui L, Bazot M, Ballester M, Darai E. Contribution of computed tomography enema and magnetic resonance imaging to diagnose multifocal and multicentric bowel lesions in patients with colorectal endometriosis. *J Minim Invasive Gynecol* 2015; 22: 776–784.
 66. Guadagno A, Grillo F, Vellone VG, Ferrero S, Fasoli A, Fiocca R, Mastracci L. Intestinal Endometriosis: mimicker of inflammatory bowel disease? *Digestion* 2015; 92: 14–21.
 67. Hudelist G, Tuttlies F, Rauter G, Pucher S, Keckstein J. Can transvaginal sonography predict infiltration depth in patients with deep infiltrating endometriosis of the rectum? *Hum Reprod* 2009; 24: 1012–1017.
 68. Benacerraf BR, Groszmann Y, Hornstein MD, Bromley B. Deep infiltrating endometriosis of the bowel wall: the comet sign. *J Ultrasound Med* 2015; 34: 537–542.
 69. Leon M, Alcazar JL. High sliding sign: a new soft marker of uterine fundus compromise in deep infiltrating endometriosis. *Ultrasound Obstet Gynecol* 2015; 45: 624.
 70. Yoon JH, Choi D, Jang KT, Kim CK, Kim H, Lee SJ, Chun HK, Lee WY, Yun SH. Deep rectosigmoid endometriosis: “mushroom cap” sign on T2-weighted MR imaging. *Abdom Imaging* 2010; 35: 726–731.
 71. Exacoustos C, Manganaro L, Zupi E. Imaging for the evaluation of endometriosis and adenomyosis. *Best Pract Res Clin Obstet Gynaecol* 2014; 28: 655–681.
 72. Ferrero S, Biscaldi E, Morotti M, Venturini PL, Remorgida V, Rollandi GA, Valenzano Menada M. Multidetector computerized tomography enteroclysis vs rectal water contrast transvaginal ultrasonography in determining the presence and extent of bowel endometriosis. *Ultrasound Obstet Gynecol* 2011; 37: 603–613.
 73. Bergamini V, Ghezzi F, Scarperi S, Raffaelli R, Cromi A, Franchi M. Preoperative assessment of intestinal endometriosis: A comparison of transvaginal sonography with water-contrast in the rectum, transrectal sonography, and barium enema. *Abdom Imaging* 2010; 35: 732–736.
 74. Reid S, Winder S, Condous G. Sonovaginography: redefining the concept of a “normal pelvis” on transvaginal ultrasound pre-laparoscopic intervention for suspected endometriosis. *Aust J Ultrasound Med* 2011; 14: 21–24.
 75. Koga K, Osuga Y, Yano T, Momoeda M, Yoshino O, Hirota Y, Kugu K, Nishii O, Tsutsumi O, Taketani Y. Characteristic images of deeply infiltrating rectosigmoid endometriosis on transvaginal and transrectal ultrasonography. *Hum Reprod* 2003; 18: 1328–1333.
 76. Pascual MA, Guerriero S, Hereter L, Barri-Soldevila P, Ajossa S, Graupera B, Rodriguez I. Diagnosis of endometriosis of the rectovaginal septum using introital three-dimensional ultrasonography. *Fertil Steril* 2010; 94: 2761–2765.
 77. Pascual MA, Guerriero S, Hereter L, Barri-Soldevila P, Ajossa S, Graupera B, Rodriguez I. Three-dimensional sonography for diagnosis of rectovaginal septum endometriosis: interobserver agreement. *J Ultrasound Med* 2013; 32: 931–935.
 78. Guerriero S, Alcazar JL, Ajossa S, Pilloni M, Melis GB. Three-dimensional sonographic characteristics of deep endometriosis. *J Ultrasound Med* 2009; 28: 1061–1066.
 79. Philip CA, Bisch C, Coulon A, de Saint-Hilaire P, Rudigoz RC, Dubernard G. Correlation between three-dimensional rectosonography and magnetic resonance imaging in the diagnosis of rectosigmoid endometriosis: a preliminary study on the first fifty cases. *Eur J Obstet Gynecol Reprod Biol* 2015; 187: 35–40.
 80. Schiffmann ML, Schafer SD, Schuring AN, Kiesel L, Sauerland C, Gotte M, Schmitz R. Importance of transvaginal ultrasound applying elastography for identifying deep infiltrating endometriosis - a feasibility study. *Ultraschall Med* 2014; 35: 561–565.
 81. Menakaya U, Reid S, Lu C, Gerges B, Infante F, Condous G. Performance of an Ultrasound Based Endometriosis Staging System (UBESS) for predicting the level of complexity of laparoscopic surgery for endometriosis. *Ultrasound Obstet Gynecol* 2016; Jan 14. DOI: 10.1002/uog.15858. [Epub ahead of print].
 82. Menakaya U, Infante F, Lu C, Phua C, Model A, Messyng F, Brainwood M, Reid S, Condous G. Interpreting the real-time dynamic ‘sliding sign’ and predicting POD obliteration: an inter-, intra-observer, diagnostic accuracy and learning curve study. *Ultrasound Obstet Gynecol* 2015; Jul 27. DOI: 10.1002/uog.15661. [Epub ahead of print].
 83. Piessens S, Healey M, Maher P, Tsaltas J, Rombauts L. Can anyone screen for deep infiltrating endometriosis with transvaginal ultrasound? *Aust N Z J Obstet Gynaecol* 2014; 54: 462–468.
 84. Tammaa A, Fritzer N, Strunk G, Krell A, Salzer H, Hudelist G. Learning curve for the detection of pouch of Douglas obliteration and deep infiltrating endometriosis of the rectum. *Hum Reprod* 2014; 29: 1199–1204.
 85. Reid S, Lu C, Casikar I, Mein B, Magotti R, Ludlow J, Benzie R, Condous G. The prediction of pouch of Douglas obliteration using offline analysis of the transvaginal ultrasound ‘sliding sign’ technique: inter- and intra-observer reproducibility. *Hum Reprod* 2013; 28: 1237–1246.
 86. Tammaa A, Fritzer N, Lozano P, Krell A, Salzer H, Salama M, Hudelist G. Interobserver agreement of non-invasive diagnosis of endometriosis by transvaginal sonography (TVS). *Ultrasound Obstet Gynecol* 2015; 46: 737–740.

SUPPORTING INFORMATION ON THE INTERNET

The following supporting information may be found in the online version of this article:

Videoclip S1 Assessment of sliding sign in an anteverted uterus, in posterior uterine fundus (a) and retrocervix (b).

Videoclip S2 Assessment of sliding sign in a retroverted uterus, in posterior uterine fundus (a) and anterior lower uterine segment (b).

Videoclip S3 Assessment of deep infiltrating endometriotic nodules in the bladder base.

Videoclip S4 Assessment of sliding sign in the uterovesical region.

Figure S1 Transvaginal sonographic images of: (a) a typical endometrioma (unilocular cyst with rather thick walls and ground glass echogenicity of cyst fluid); and (b) an atypical endometrioma (irregular, apparently solid, hyperechogenic structure within the cystic lesion, with ground glass echogenicity of cyst fluid).

Figure S2 Ultrasound image demonstrating ‘kissing’ ovaries sign: two ovaries containing endometriomas are fixed to each other by adhesions in the pouch of Douglas.

Figure S3 Color Doppler ultrasound image demonstrating decidualization of endometrioma in pregnancy.

Figure S4 Schematic drawings demonstrating deep infiltrating endometriosis located in: (a) trigonal zone of bladder; (b) base of bladder; (c) dome of bladder; (d) extra-abdominal bladder.

Figure S5 Schematic drawing and ultrasound image demonstrating most frequent location of endometriotic bladder nodules: bladder base.

Figure S6 Schematic drawing demonstrating location of uterovesical obliteration. If the bladder does not slide freely over the anterior uterine wall, then the ‘sliding sign’ is negative and the uterovesical region is classified as obliterated.

Figure S7 (a) Schematic drawing demonstrating measurement of distance from ureteric orifice to level of stricture, necessary in cases of deep infiltrating endometriosis (DIE) in the ureters. (b) Ultrasound image showing distal ureter obstructed by DIE; proximal hydronephrosis is seen.

Figure S8 (a) Schematic drawings and ultrasound images demonstrating location and different morphological appearances of deep infiltrating endometriosis (DIE) in posterior vaginal fornix. A normal ultrasound image is shown for comparison. (b) Ultrasound image demonstrating forniceal lesion, characterized by increased thickness of posterior vaginal fornix above line passing along lower border of posterior lip of cervix (under peritoneum).

Figure S9 Schematic drawing (a) and ultrasound image (b) demonstrating ‘diabolo-like’ nodule of deep infiltrating endometriosis in posterior vaginal fornix extending into anterior rectal wall.

Figure S10 Ultrasound images and schematic drawings demonstrating deep infiltrating endometriosis (DIE) of the uterosacral ligaments (USL). (a) Longitudinal and transverse planes: normal USLs are usually not visible on ultrasound. (b) Location of USL DIE in the mid-sagittal view. (c) Location of DIE in the right USL in transverse view. (d) Location of USL DIE at the torus uterinus in a transverse view through the pelvis.

Figure S11 Schematic drawings and transvaginal ultrasound image demonstrating deep infiltrating endometriotic (DIE) rectal lesions. (a) DIE in the upper anterior rectum. (b) Multifocal DIE in the rectum.

Figure S12 Schematic image showing histological layers of normal rectum (a), with corresponding layers shown in ultrasound image of a bowel loop with nodule of deep infiltrating endometriosis in the bowel wall (b).

Figure S13 (a–c) Schematic drawings and corresponding ultrasound images demonstrating different levels of obliteration of the pouch of Douglas (POD) in an anteverted uterus: (a) retrocervical level; the ‘comet’ sign is also present; (b) mid-posterior uterus; the ‘Indian headdress’ (or ‘moose antler’) and ‘pulling sleeve’ signs are also present; (c) posterior uterine fundus; the pulling sleeve sign is also present. (d) POD obliteration in a retroverted uterus.

Figure S14 Schematic drawing and ultrasound image demonstrating ‘mushroom cap’ sign. Retraction within nodule of deep infiltrating endometriosis in rectosigmoid results in overestimation of true thickness of lesion and, in some cases, in underestimation of true length of bowel segment affected by lesion.

Figure S15 Schematic drawing and ultrasound image demonstrating measurement of distance from anal verge to deep infiltrating endometriotic lesion of bowel.

Figure S16 Doppler images demonstrating rectal cancer with marked vascularity.

Figure S17 Ultrasound image obtained during rectal water contrast transvaginal ultrasound in a woman with deep infiltrating endometriosis.

Figure S18 Ultrasound image obtained during gel sonovaginography in a woman without endometriosis, showing pouch of Douglas.

Figure S19 Multiplanar view of pelvic floor as visualized by introital ultrasound with three-dimensional reconstruction with render mode, showing endometriotic nodule in rectovaginal septum, between rectum and vagina.

Figure S20 Transvaginal elastogram of nodule of rectal deep infiltrating endometriosis; nodule has mainly high stiffness, with some low-stiffness spots.



Enfoque sistemático para la evaluación ecográfica de la pelvis en mujeres con posible endometriosis, incluyendo términos, definiciones y mediciones: una opinión consensuada del Grupo Internacional de Análisis de la Endometriosis Profunda

RESUMEN

La declaración del Grupo Internacional de Análisis de la Endometriosis Profunda (IDEA, por sus siglas en inglés) es una opinión basada en un consenso sobre los términos, definiciones y medidas que se pueden utilizar para describir las características ecográficas de los distintos fenotipos de la endometriosis. Actualmente es difícil comparar los resultados entre los estudios publicados porque los autores utilizan términos diferentes para describir las mismas estructuras y localizaciones anatómicas. Esperamos que los términos y definiciones propuestas en este documento se adopten en centros de investigación de todo el mundo. Esto resultaría en un uso uniforme de la nomenclatura para describir la ubicación y el alcance de la endometriosis en la evaluación ecográfica. Creemos que la normalización de la terminología permitirá realizar comparaciones significativas entre futuros estudios de mujeres con diagnóstico de endometriosis mediante ecografía y debería facilitar la investigación entre múltiples centros de investigación.

疑似子宫内膜异位症的女性中超声评估盆腔的系统方法（包括术语、定义和测量方法）：国际深部浸润型子宫内膜异位症分析 (IDEA) 研究组达成的一致共识

国际深部浸润型子宫内膜异位症分析研究组 (International Deep Endometriosis Analysis group, IDEA) 发表声明，对用于描述子宫内膜异位症不同表型的超声特点的术语、定义和测量方法达成一致意见。目前很难对已发表的研究结果进行比较，因为作者在描述相同结构和解剖部位时采用不同的术语。我们希望本文提出的术语和定义能够被全世界的医疗中心所采用，从而在描述子宫内膜异位症超声部位和程度时采用一致的术语。我们相信，术语的标准化将使我们今后对超声诊断为子宫内膜异位症的女性进行研究时能够进行有意义的比较，并且促进多中心研究。